



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JANUARY 2024**

Course : EC3105 (C Programming)

Time : 8.00AM – 11.00AM
(3 hours)

Lecturer : Dr. Nur Inshirah Naqiah

Date : 20 May 2024

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **3** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

- 1 a. ✓ What operating system does Linux use? (6 MARKS)
- b. ✓ What is the difference between `$(localhost:~$)` and `$(localhost:~#)` signs in a Linux environment? (9 MARKS)
- c. ✓ State appropriate Linux command(s) to answer the following questions: (10 MARKS)
- i. Copy MyFile.txt to YourFile.txt
 - ii. Create a new file HerFile.txt and type its content "I love C Programming".
 - iii. Display the content of HerFile.txt
 - iv. Copy HerFile.txt to HisFile.txt
 - v. What is the content of HisFile.txt?
- 2 a. ✓ Develop a program employing symbolic constant `#define` to determine the circumference and area of a circle. (5 MARKS)
- b. ✓ Develop a program using an IF-ELSE statement to ascertain the largest among three numbers through the following two-step procedure: (10 MARKS)
- i. Compare the initial number with the second number and choose the greater of the two;
 - ii. Compare the previously selected number from step (i) with the third number and choose the greater one;
- c. ✓ Develop a C program to identify the largest among three numbers provided by the user using Nested IF-ELSE statements. (5 MARKS)
- d. ✓ Develop a program utilizing a switch statement to output the day of the week based on the value of an integer variable 'day', where Sunday is represented by day 1. (5 MARKS)

3 a. Create a program that repeatedly doubles the number 2 (2, 4, 8, 16, ...) and prints the resulting numbers if they are less than 100 by using a WHILE loop. **(8 MARKS)**

b. Create a program to find the largest of two numbers using functions. (functions with arguments but no return values) **(5 MARKS)**

c. Write a function to print your name, as shown below: **(8 MARKS)**

```
*
*      Your Name Here      *
*
```

Write a call as it would be coded in a calling function, such as main.

d. The program below contains syntax errors which will be reported during compilation. List the errors: **(4 MARKS)**

```
*/This program below contains MANY syntax errors/*
include(stdio.h);
main()
{
    float num
    num:=305.68;
    printf("num is")
    printf('%6.1f\n',Num);
}
```

- 4 a. Employ a for loop to accumulate the values stored in an integer array named 'numbers', consisting of five elements. Save the cumulative result in an integer variable named 'total'. **(10 MARKS)**
- b. Utilize a printf statement to display the third element of an integer array named 'totals'. **(6 MARKS)**
- c. Write a program to open a file named INVENTORY and store in the following data: **(9 MARKS)**

Item name	Number	Price	Quantity
Z1	242	15.50	112
Z2	344	32.40	45
Z3	511	28.30	128

The program is supposed to read this data from the file INVENTORY and display the table with the value of each item.

-END OF QUESTION PAPER-